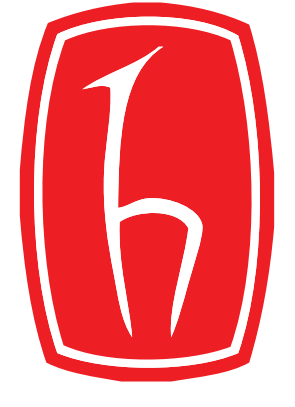




OASIS

AI-Assisted Mobile Reporting and
VR Visualization for Disaster Management



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WHY OASIS?

Official teams arrive **late** to the field



Near real-time data is collected by turning
citizens into sensors

Reliability of raw reports is low



LLM and VLM models analyze reports and
produce **deterministic severity scores**

GPS inconsistencies cannot be detected



Location is **cross-validated** via EXIF

Scattered reports lack meaning



Spatial clustering is performed via **DBSCAN**

Vertical disasters cannot be perceived on
2D maps



Realistically visualized via 3D GIS interface and VR

MOBILE INTERFACE



WEB INTERFACE



SPATIAL CLUSTERING OF DISASTER REPORTS

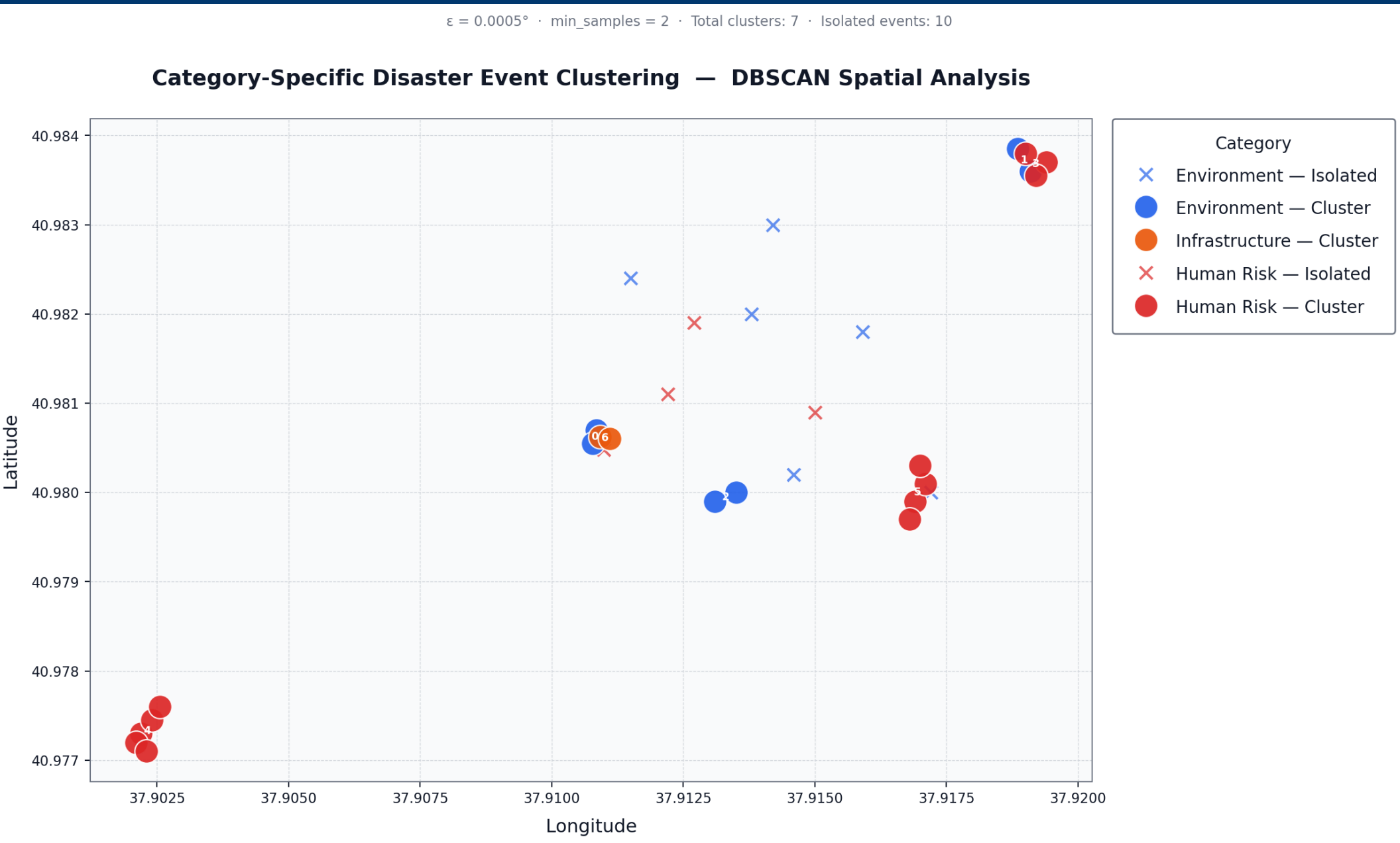


Fig. 3 — DBSCAN output on 50 synthetic reports across Akyazı district, Ordu, Turkey

1. Group reports → by disaster category
2. Per group → compute pairwise distances
3. For each point → find neighbours within $\epsilon = 0.0005^\circ$ (~55 m)
4. neighbours $\geq$ minPts (3) → point labelled core point
5. Core points within ϵ → merged into same cluster
6. Non-core but within ϵ of core → border point, joins cluster
7. Remaining points → noise (isolated event, cluster = -1)

Scattered reports are geo-clustered by **proximity**
and **disaster type**, enabling near real-time
situational awareness for emergency responders.

ALGORITHMS

DBSCAN Spatial Clustering

$\epsilon = 0.0005^\circ$ (~55m), minPts = 3

$$RegionQuery(P, \epsilon) = \{Q \in D \mid dist(P, Q) < \epsilon\}$$

Haversine Distance

$$a = \sin^2\left(\frac{\Delta lat}{2}\right) + \cos(lat_1) \cdot \cos(lat_2) \cdot \sin^2\left(\frac{\Delta lon}{2}\right)$$

$$d = 2R \cdot \arctan2(\sqrt{a}, \sqrt{1-a}), \quad R = 6,371,000 \text{ m}$$

Deterministic Severity Score

$$S = \sum_k \min\left(\sum_i w_i \cdot b_i, cap_k\right), \quad b_i \in \{0,1\}, S \in [1,10]$$

$K = \{human(4.0), infrastructure(3.0), environment(2.0), communication(1.0)\}$

Reliability Score

$$R = 0.5 + 0.5 \times \frac{N_{cross}}{N_{active}}, \quad R \in [0.30, 1.00]$$

Dynamic Cluster Buffer

$$r = \min(\max(d_{max} \times 1.2, r_{min}), r_{max})$$

$$d = 2R \arctan\left(\sqrt{\frac{a}{1-a}}\right)$$

$$a = \sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos\varphi_1 \cos\varphi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)$$

EXIF GPS Conversion

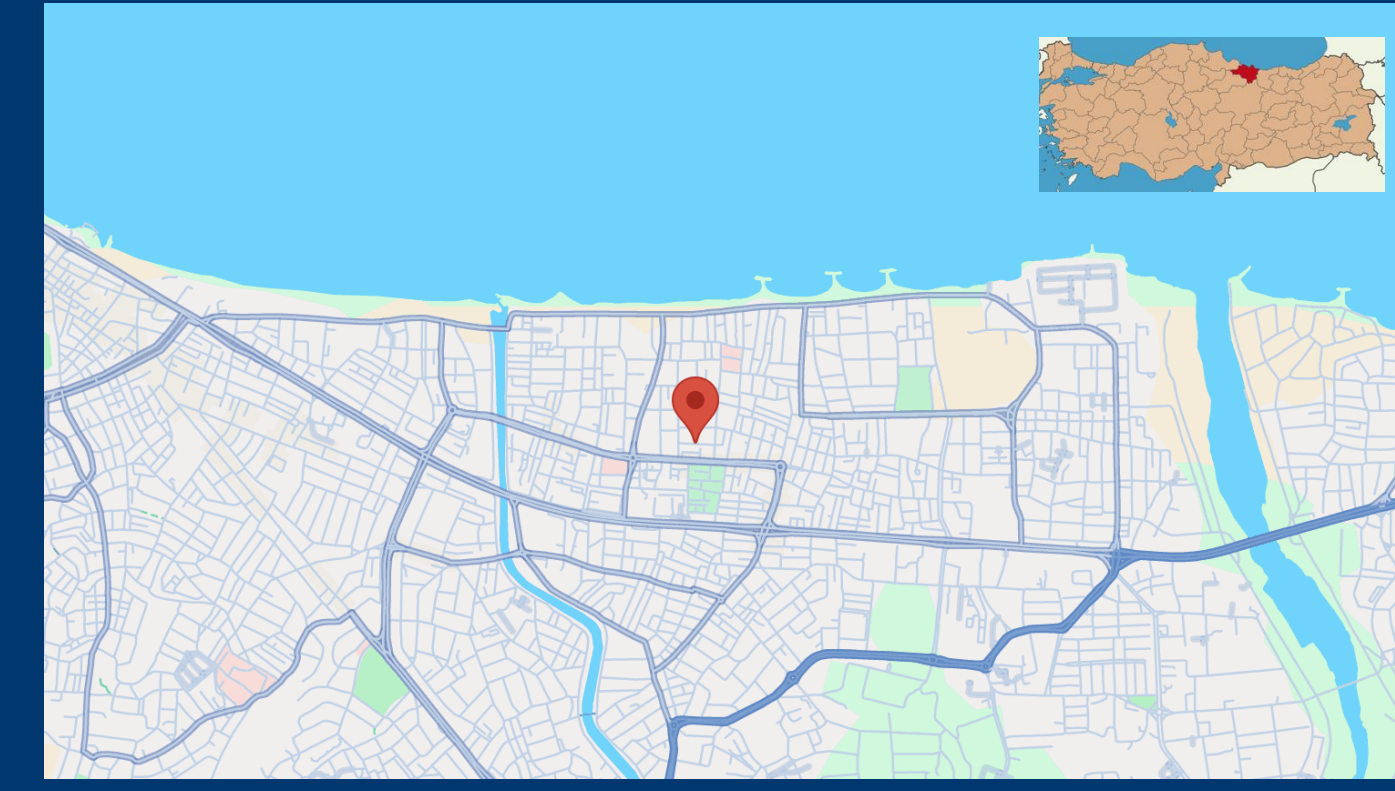
$$DD = D + \frac{M}{60} + \frac{S}{3600}$$

$$|d_{exif} - d_{reported}| > \text{threshold} \rightarrow \text{location warning}$$

Severity weighting adapted from Sendai Framework (UNDRR, 2015–2030)

Reliability model based on VGI quality assessment — Goodchild (2007)

PILOT AREA - AKYAZI, ORDU

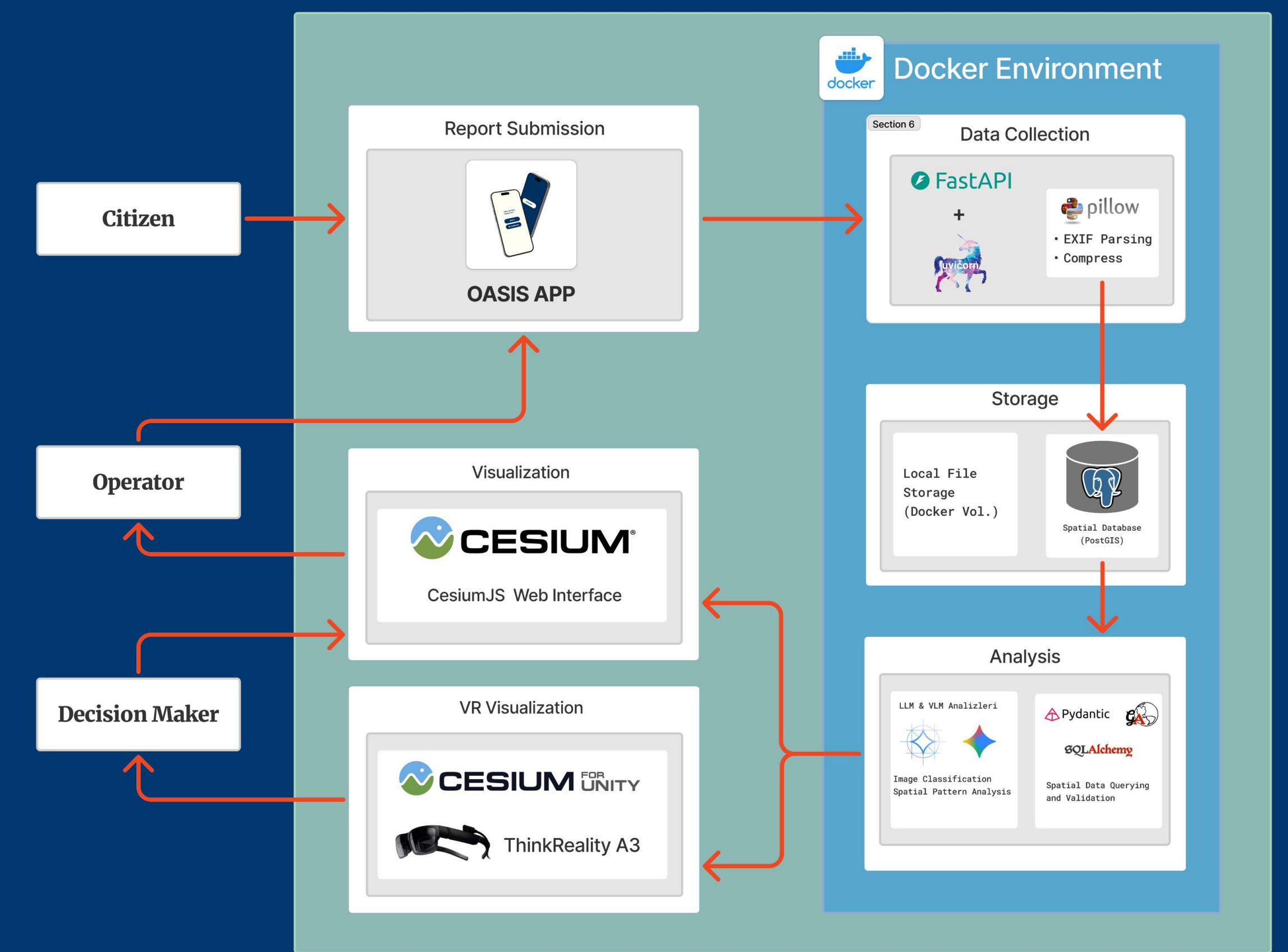


Akyazı, Altınordu/Ordu is a
flood-prone urban
neighborhood selected for its
high historical flood frequency
and dense residential exposure.

DATA FLOW



SYSTEM ARCHITECTURE



MODEL COMPARISON: Gemma 3 27B vs spaCy-Tr

	spaCy-Tr	Gemma 3 27B
Type	Rules-based NLP	Cloud-based LLM
Accuracy (50 reports)	78%	86%
Turkish Support	Requires training	Built-in
Token Cost	Low	High

Gemma 3 27B preferred: 86% accuracy on disaster type &
severity categorization (deterministic label matching, 50 reports),
with zero domain-specific training.

Note: If spaCy is trained on disaster data, **cost advantage** may
become significant.

FUTURE PLANS



Non-disaster
field testing



Speed and cost
optimization



VR training

